Retrieval Interference in Sentence Comprehension: A new theory and data

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Cue-based Retrieval Interference

Similarity-based Retrieval Interference in sentence processing occurs when the retrieval of a constituent from memory is interfered with by another, similar, constituent. Van Dyke & McElree (2006) have shown that while there are no reading time differences between (1a) and (1b) *per se*, readers slow down at *fixed* but not at *sailed* if they have to memorize a list of fixable but not sailable nouns prior to reading the sentence. Fixable nouns in the memory list seem to interfere with the retrieval of *the boat* in (1a) but not in (1b).

(1) It was ***the boat*** that the guy who lived by the sea ...
a. .... **fixed** in two sunny days.
b. .... **sailed** in two sunny days.

This effect is explained in terms of cue-overload caused by competitors.

The feature [+fixable] which *the boat* has and which is used by *fixed* in (1a) for retrieving it is overloaded by the nouns in the memory list, while [+sailable] in (1b) is not - thus the slowdown in (1a) given a memory list of fixable but not sailable nouns.
The slowdown is a consequence of lower retrieval probability in a cue-overload situation as predicted by the SAM Model (Gillund & Shiffrin, 1984) formalized below for a target item I_i, a set of retrieval cues Q_j and the strength of association $S(Q_j, I_i)$ between item and cues.

$$P(I_i|Q_1, ..., Q_m) = \frac{\prod_{j=1}^{m} S(Q_j, I_i)^{w_j}}{\sum_{k=1}^{N} \prod_{j=1}^{m} S(Q_j, I_k)^{w_j}}$$

The probability of a successful retrieval thus depends on the match between cues and item **in relation** to the matches between cues and all items in memory, i.e. the competitors in the memory list as well.

Experiments 1 and 2

One prediction of this model is that features not forming part of the cues should not lead to interference. Another is that increasing similiarity should lead to increasing confusability and thus to increasing processing times.

We tested these predictions in two experiments. We manipulated the match of two NPs in gender (masculine/feminine) and NP-type (nationality/occupation) yielding 4 conditions: (a) full match, (b) gender match, (c) NP-type match and (d) no match.

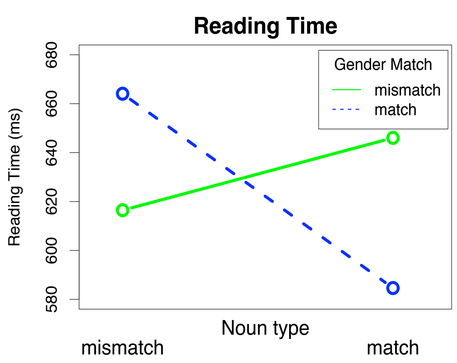
(2)

Der Polizist *the policewoman.MASC*	hat den Zaun, [den *has the fence, [that*	der Nachtwächter *the night guard.MASC*	deutlich **sah**,] sehr ärgerlicherweise übersehen. *clearly* ***saw**,] very unfortunately overseen.*
Die Polizistin *the policeman.FEM*		die Nachtwächterin *the night guard.FEM*	
Der Spanier *the spaniard.MASC*		der Italiener *the italian.MASC*	
Die Spanierin *the spaniard.FEM*		die Italienerin *the italian.FEM*	

If retrieval cues are an inherent part of the retrieval site, there should be no interference effect for sentences like (2), because a match of two NPs in certain properties is not a sensible retrieval cue. On the other hand, if retrieval cues can be provided by the context, i.e. the target itself, we expect an interference effect, which should be stronger for (a) than for (b) and (c), with (d) serving as a baseline.

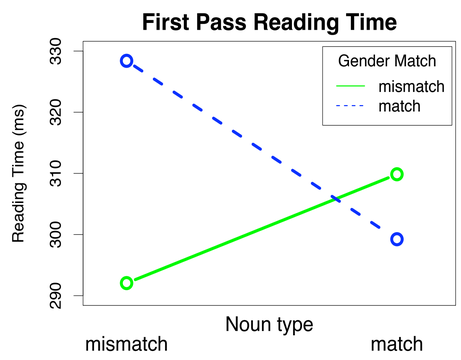
Experiment 1 (Self-paced Reading)

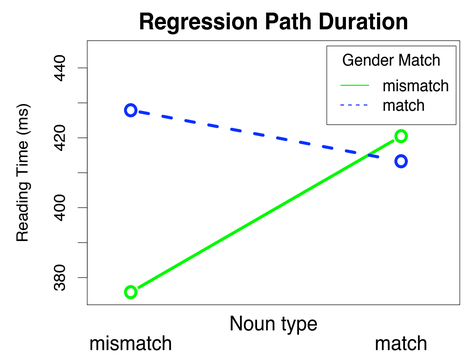
Results: A multilevel model on log-transformed reading times for trials with correctly answered questions revealed a significant interaction at the embedded verb **sah** in FPRT (t=-2.59) with no main effects (|t|<1.3). On the preceding region, we found a main effect of noun type (t=-2.34), but no interaction (t=-0.77).



Results: A multilevel model on log-transformed reading times for trials with correctly answered questions revealed a significant interaction at the embedded verb **sah** in FPRT (t=-2.31), RPD (t=-2.22) and TFT (t=-2.74). There were no significant effects on the preceding region (|t|<1.2).

Experiment 2 (Eye Tracking)



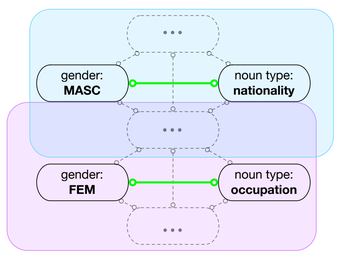


Conclusion

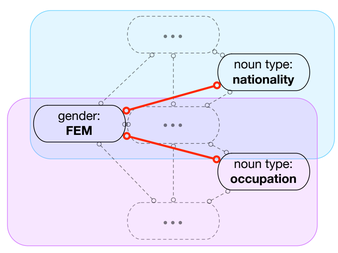
Our results contradict both predictions. First, there are interference effects, reflected in longer RTs. Second, interference does not increase with increasing similarity.
In sum, our results cannot be explained for by Van Dyke & McElree's account of interference.

Conflicting Bindings Retrieval Interference

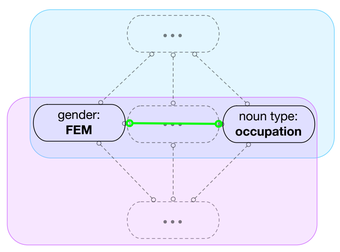
We assume that objects in memory (and thus NPs) are feature bundles pairwise bound to each other via links we call bindings. Every object is uniquely identified by the sum of its bindings. The fact that one matching feature causes interference while two matching features lead to no or negative interference can be explained under the assumption that retrieval intereference is not caused by matching features, but conflicting bindings. We call two bindings (between e.g. gender and noun type) conflicting if they bind one value of one feature to different values of the other feature.



A. No Match



B. One Match



C. Full Match

If the NPs mismatch in gender and noun type, the bindings involving these features will be non-conflicting (cf. Panel A) and processing will be relatively easy. If the NPs match in exactly one feature (e.g. noun type in Panel B), bindings involving these features will be conflicting, thus causing interference due to more conflicting bindings, which results in longer RTs. If the NPs match in both features, there will be no conflicting bindings since bindings involving the experimentally relevant features will be identical for both NPs and thus not conflicting - making processing relatively easy.

We call this sort of intereference *Conflicting Bindings Retrieval Interference (CBRI)*. CBRI, based on the assumption that the number of conflicting bindings is the major determinant of interference strength can thus explain our experimental results, while Van Dyke and McElree's account cannot. Furthermore, CBRI can explain Van Dyke and McElree's results under the assumption that conflicting bindings involving features that are part of the retrieval cues and therefore more salient cause stronger interference.

References
Van Dyke and McElree (2006). Retrieval Interference in Sentence Comprehension. Journal of Memory and Language, 55, 157-166